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journal homepage: www.elsevier.com/locate/jmonecoResolving the missing deflation puzzle[☆]Martín Harding^a, Jesper Lindé^{b,c}, Mathias Trabandt^{d,e,*}^a Bank of Canada, 234 Wellington Street, Ottawa, ON K1A 0G9, Canada^b IMF (Monetary and Capital Markets Department), Washington DC, United States^c CEPR, London, United Kingdom^d Goethe University Frankfurt, Theodor-W.-Adorno-Platz 3, Frankfurt am Main 60323, Germany^e Halle Institute for Economic Research (IWH), Germany

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ABSTRACT

A resolution of the missing deflation puzzle is proposed. Our resolution stresses the importance of nonlinearities in price- and wage-setting when the economy is exposed to large shocks. We show that a nonlinear macroeconomic model with real rigidities resolves the missing deflation puzzle, while a linearized version of the same underlying nonlinear model fails to do so. In addition, our nonlinear model reproduces the skewness of inflation and other macroeconomic variables observed in post-war U.S. data. All told, our results caution against the common practice of using linearized models to study inflation and output dynamics.

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* Corresponding author at: Goethe University Frankfurt, Theodor-W.-Adorno-Platz 3, Frankfurt am Main 60323, Germany.

E-mail addresses: mharding@bankofcanada.ca (M. Harding), jlindel@imf.org (J. Lindé), mathias.trabandt@gmail.com (M. Trabandt).

1. Introduction

The Great Recession has left macroeconomists with many puzzles. One such puzzle is the alleged breakdown of the relationship between inflation and the output gap – also known as the Phillips curve. The Great Recession generated an extraordinary decline in U.S. GDP of about 10 percent relative to its pre-crisis trend while inflation dropped only by about 1.5 percent (see, e.g. [Christiano et al., 2015](#)). The modest decline in inflation was surprising to many macroeconomists. For instance, New York Fed President John C. Williams (2010, p. 8) wrote: “The surprise [about inflation] is that it’s fallen so little, given the depth and duration of the recent downturn. Based on the experience of past severe recessions, I would have expected inflation to fall by twice as much as it has”.

The small drop in inflation has been referred to as the “missing deflation puzzle” and has attracted considerable interest among academics and policymakers. [Hall \(2011\)](#) argues that since inflation fell so little in the face of the large contraction in GDP, one might view inflation as being essentially exogenous to the economy. Specifically, [Hall \(2011\)](#) argues that popular DSGE models based on the simple New Keynesian Phillips curve “cannot explain the stabilization of inflation at positive rates in the presence of long-lasting slack”. Similarly, [Ball and Mazumder \(2011\)](#) argue that Phillips curves estimated for the post-war pre-crisis period in the United States cannot explain the modest fall in inflation during the years of the financial crisis from 2008 through 2010. They argue that the fit of the standard Phillips curve deteriorates sharply during the crisis. A further challenge to the New Keynesian Phillips curve is raised by [King and Watson \(2012\)](#), who find a large discrepancy between actual inflation and inflation predicted by the workhorse [Smets and Wouters \(2007\)](#) model. Interestingly, all above contributions use linearized variants of the New Keynesian Phillips curve in their analysis.

Our proposed resolution of the missing deflation puzzle rests on two key elements. First, it is key to introduce real rigidities in price- and wage-setting. To do this, we follow [Dotsey and King \(2005\)](#) and [Smets and Wouters \(2007\)](#) and use the [Kimball \(1995\)](#) aggregator instead of the standard [Dixit and Stiglitz \(1977\)](#) aggregator. The Kimball aggregator introduces additional strategic complementarities in the price- and wage-setting, which lowers the sensitivity of prices and wages to the relevant wedges for a given degree of price- and wage-stickiness. As such, the Kimball aggregator is commonly used in New Keynesian models, see e.g. [Smets and Wouters \(2007\)](#), as it allows to simultaneously account for the *macroeconomic evidence* of a low Phillips curve slope and the *microeconomic evidence* of frequent price changes.

Second, the standard procedure of linearizing all equilibrium equations around the steady state, except for the zero lower bound (ZLB) constraint on the nominal interest rate, introduces large approximation errors when large shocks hit the economy as was the case during the Great Recession. Implicit in the linearization procedure is the assumption that the linearized solution is accurate even far away from the steady state. Our analysis shows that the linearized solution is very inaccurate far away from the steady state. We show that one ought to use the nonlinear model solution instead of the linearized model solution to understand the output-inflation dynamics during the Great Recession. The nonlinearity implied by the Kimball aggregator is a key model feature that accounts for the differences between the linearized and nonlinear model solutions: it is crucial to resolve the missing deflation puzzle. The Kimball aggregator implies that the demand elasticity for intermediate goods is state-dependent, i.e. firms’ demand elasticity is an increasing function of its relative price and the demand curve is quasi-kinked.¹ While the fully nonlinear model takes the state-dependence of the quasi-kinked demand curve explicitly into account, a linear approximation replaces this key nonlinearity by a linear function. When the economy is exposed to large shocks the state-dependence of the quasi-kinked demand curve becomes quantitatively important and the linear approximation ceases to provide accurate results.

A key contribution of our work is that we provide a structural general equilibrium model which can account for the alleged breakdown of the Phillips curve during the Great Recession.² That is, a nonlinear formulation of our model generates a very modest fall of inflation in the wake of a large and persistent contraction of output. By contrast, the linearized version of the same underlying nonlinear model predicts a much larger fall of inflation and thereby fails to account for the output-inflation dynamics in the data. Therefore, our model provides a resolution to the missing deflation puzzle: the puzzle arises only if one uses the linearized solution to predict inflation in the Great Recession. [Section 3.1.2](#) provides the intuition and [Fig. 2](#) illustrates this key result graphically. In a nutshell, our model implies nonlinear price and wage Phillips curves, i.e. nonlinear relationships between price and wage inflation and the output gap. The slopes of our price and wage Phillips curves are notably flatter in recessions than in booms, i.e. our Phillips curves have a banana- or boomerang-shape as in the seminal paper by [Phillips \(1958\)](#). Consequently and consistent with the data, our nonlinear model implies that inflation falls relatively little in deep recessions. By contrast, the linearized model erroneously predicts a much larger fall in inflation by extrapolating local decision rules around the steady state to deep recessions far away from the steady state.³

We establish our main results in a variant of the [Erceg et al. \(2000\)](#) (EHL henceforth) benchmark model. The EHL model, however, does not allow for endogenous capital accumulation and other real rigidities like habit formation in consumer preferences and investment adjustment costs. Therefore, we examine the robustness of our results in an estimated medium-sized New Keynesian model which include the features listed above. Specifically, we use the benchmark

¹ See [Dupraz \(2017\)](#) for a microfoundation of a kinked demand curve theory.

² Thus, we provide a structural interpretation of the emerging body of empirical literature that provides macroevidence in favor of nonlinearities in the Phillips curve, see for instance [Doser et al. \(2018\)](#), [Gagnon and Collins \(2019\)](#) and the references therein.

³ Recent work by [Boneva et al. \(2016\)](#) and [Lindé and Trabandt \(2018\)](#) suggests that linearization can also produce misleading results about the effects of fiscal policy.

Smets and Wouters (2007, SW henceforth), model, which is a multishock version of the seminal model of Christiano et al. (2005). We show that a re-estimated version of the SW model which gives a more prominent role to Kimball (1995) kinked demand – similar to our parameterization in the EHL model – resolves the missing deflation puzzle and hence does not rely on markup shocks to avoid deflation during the Great Recession and its aftermath.

Moreover, the insights from the EHL model carry over to the nonlinear SW model with a more prominent role for kinked demand. By performing stochastic simulations with both models, two further important insights are also established.

First, a large body of recent work – see for instance IMF (2016, 2017) – has been aimed towards understanding the absence of upward pressure of price and wage inflation during the recovery from the Great Recession, also called the missing inflation puzzle. The nonlinear formulation of our model offers an explanation for this phenomenon. Given the nonlinear Phillips curves in our model, price and wage inflation remain subdued until the level of economic activity has recovered sufficiently relative to its potential. Put differently, even though economic growth may resume after a deep recession, price and wage inflation will only increase modestly until economic slack has subsided sufficiently. Therefore, our model does not only resolve the missing deflation puzzle, it also offers an explanation for the missing inflation puzzle. According to our model, subdued price and wage inflationary pressures as observed during the recovery from the Great Recession are indicative of substantial economic slack.⁴ While the linearized model suggests that price and wage inflation should rise sharply during a recovery from a deep recession, the nonlinear model suggests otherwise: price and wage inflation will not rise until economic slack has narrowed sufficiently from its elevated recession level.

Second, our nonlinear model can be used to understand the positive skewness in post-war U.S. price inflation, i.e. that price inflation scares are much more common than deflationary episodes. We establish this result by comparing the skewness for inflation in the data and our model. Consistent with the data, our nonlinear model implies positive skewness in inflation. In addition, our nonlinear model also generates positive skewness for wage inflation and thereby Kimball aggregation in the labor market provides scope to microfound downward nominal wage rigidity. By contrast, the linearized version of the model fails to account for the skewness of these time series. All told, our nonlinear model allows us to simultaneously explain the quantitative inflation performance prior to the Great Recession, the missing deflation during the Great Recession, as well as the missing inflation in recent years.

The remainder of the paper is organized as follows. Section 2 discusses related literature on micro- and macroeconomic evidence. Section 3 presents the stylized New Keynesian model with stickiness and real rigidities in price- and wage-setting. Section 4 discusses the results based on the stylized model. Section 5 examines the robustness of our results in an estimated medium-sized New Keynesian model. Finally, Section 6 provides concluding remarks.

2. Related literature and evidence

In this section, we discuss literature that is related to our work. We also discuss micro evidence that sheds light on the empirical relevance of real rigidities with a special emphasis on Kimball versus Dixit–Stiglitz aggregation. Finally, we also discuss macroeconomic evidence related to the missing deflation puzzle.

2.1. Micro evidence

A large body of literature has documented patterns of nominal price stickiness at a disaggregated good level, see e.g. Klenow and Malin (2010); Nakamura and Steinsson (2008) and the references therein. The empirical findings suggest that prices change on average about every three to four quarters, i.e. nominal stickiness is relatively short-lived. Christiano et al. (1999) and Gertler and Karadi (2015) have documented persistent real effects of monetary policy shocks on output. Short-lived price stickiness and persistent real effects of monetary shocks can be reconciled with each other in modeling frameworks with sufficient real rigidities, i.e. features that make firms reluctant to change their prices by large amounts.⁵

Amid this background, an empirical literature has started to emerge recently to examine whether there are quantitatively important real rigidities present in the data, see e.g. Dossche et al. (2010); Klenow and Willis (2016); Lein (2015), and Levin and Yun (2008).⁶ In general, the empirical literature appears to provide evidence in favor of Kimball aggregation rather than for Dixit–Stiglitz aggregation. That is, the empirical literature has estimated demand curves which exhibit state-dependent demand elasticities rather than constant demand elasticities. Put differently, there appears to be more support for quasi-kinked demand curves as implied by Kimball aggregation than for log-linear demand as implied by Dixit–Stiglitz aggregation.

However, although the evidence provided by the empirical literature supports Kimball over Dixit–Stiglitz aggregation in general, a balanced assessment of the evidence provided by the above papers is that it does not seemingly support the degree of real rigidities needed to fit the macro evidence. That said, the empirical literature has typically used micro data

⁴ See e.g. Erceg and Levin (2014) who argue that economic slack was bigger than commonly thought following the Great Recession.

⁵ There is a long-standing I/O literature which consistent with the implications of our nonlinear model argues that prices tend to respond faster in booms than in recessions, see for instance Peltzman (2000) and the references therein.

⁶ There has also been considerable theoretical work that explores the effects of real rigidities in firms' price-setting behavior for small shocks, see e.g. Dotsey et al. (1999); Dotsey and King (2005); Dupraz (2017); Kocherlakota (2017) and Kleshchelski and Vincent (2009). In this section, however, we focus on micro evidence on real rigidities.

which does not cover the Great Recession period which is at the core of our study. In addition, real rigidities are difficult to identify and difficult to measure in the data as noted by [Gopinath and Itskhoki \(2010\)](#). Aggregate real rigidities imply a sluggish response of firms' marginal costs to aggregate shocks. Firm-level real rigidities imply a muted response of firms' prices conditional on price adjustment to marginal cost shocks. Since data on marginal costs is usually unavailable, it is hard to test these mechanisms directly. So, the evidence provided in e.g. [Klenow and Willis \(2016\)](#) is merely indicative, and their implied large idiosyncratic shock variance needed for their findings to be compatible with our estimate of the demand elasticity may be reconciled by allowing for firm-specific customer relationships as in [Levin and Yun \(2008\)](#) and [Kleshchelski and Vincent \(2009\)](#).

2.2. Macro evidence

Recent research has examined possible resolutions of the missing deflation puzzle. Using standard time series methods on U.S. data, an emerging body of literature finds support for the presence of nonlinearities in the Phillips curve. [Doser et al. \(2018\)](#) estimate a two-state regime Phillips curve and report a lower slope of the Phillips curve in a state in which the unemployment rate is elevated. [Gagnon and Collins \(2019\)](#) argue – based on time series evidence for a wide range of inflation time series – that the Phillips curve is likely to be highly nonlinear when inflation is low. The evidence provided in both papers is consistent with our structural model.

Another body of literature has used structural models to examine possible explanations for the missing deflation puzzle. [Lindé et al. \(2016\)](#) and [Fratto and Uhlig \(2018\)](#) find that the [Smets and Wouters \(2007\)](#) model relies on large offsetting positive price markup shocks to cope with the small drop in inflation in the face of a persistent fall in output observed during the Great Recession.

Recent research has also emphasized that financial frictions can be helpful to account for a small elasticity between output and inflation witnessed during the Great Recession. [Christiano et al. \(2015\)](#) use a model to show that the observed fall in total factor productivity and the rise in firms' cost to borrow funds for working capital played critical roles in accounting for the small drop in inflation that occurred during the Great Recession. [Del Negro et al. \(2015\)](#) show that the introduction of a financial accelerator together with a flattening of the Phillips curve can account for the small drop in inflation in the Great Recession. [Gilchrist et al. \(2017\)](#) develop a model in which firms face financial frictions when setting prices in an environment with customer markets. Financial distortions create an incentive for financially constrained firms to raise prices in response to adverse financial or demand shocks in order to preserve internal liquidity and avoid accessing external finance. While financially unconstrained firms cut prices in response to these adverse shocks, the share of financially constrained firms is sufficiently large in their model to attenuate the fall in inflation in response to a large contraction in GDP. [Gilchrist et al. \(2017\)](#) examine a micro data set which supports the implications of their model.

Our work is also related to [Aruoba et al. \(2017\)](#). Using asymmetric price and wage adjustment costs, the authors show that their model can produce skewness in inflation and output growth as observed in the data. Our work differs from [Aruoba et al. \(2017\)](#) in at least two ways. First, we use the [Kimball \(1995\)](#) aggregator to generate skewness in inflation and other macro data. Second, our analysis focuses on the interplay of large shocks and the zero lower bound to characterize nonlinearities in price and wage Phillips curves.

More generally, our work offers an alternative and perhaps complementary explanation to understand the missing deflation puzzle. Our resolution of the puzzle stresses the nonlinear influence of strategic complementarities and real rigidities in price- and wage-setting. We find it attractive due to its simplicity and due to its ability to address additional issues beyond the missing deflation puzzle which are discussed in the following sections as well as in the conclusion.

3. A stylized new Keynesian model

The simple model we use is very similar to the EHL model with gradual price and wage adjustment. It differs from EHL in two ways. First, by allowing for [Kimball \(1995\)](#) aggregators in price- and wage-setting (with the standard [Dixit and Stiglitz, 1977](#) specification used by EHL as a special case). Second, by including a discount factor shock (or more generally a savings shock) into the model.

3.1. Model

This section summarizes the key elements of the model. The complete specification of the nonlinear and linearized formulation of the model is provided in the Technical Appendix available at the authors' websites.

3.1.1. Firms and price setting

Final goods production Competitive firms use intermediate goods $Y_{f,t}$ to produce final goods Y_t using the production technology $\int_0^1 G_Y\left(\frac{Y_{f,t}}{Y_t}\right) df = 1$. Following [Dotsey and King \(2005\)](#) and [Levin et al. \(2007\)](#), we assume that $G_Y(\cdot)$ is given by the following concave and increasing function:

$$G_Y\left(\frac{Y_{f,t}}{Y_t}\right) = \frac{\omega_p}{1 + \psi_p} \left((1 + \psi_p) \frac{Y_{f,t}}{Y_t} - \psi_p \right)^{\frac{1}{\omega_p}} + 1 - \frac{\omega_p}{1 + \psi_p} \quad (1)$$

where $\omega_p = \frac{\phi_p(1+\psi_p)}{1+\phi_p\psi_p}$. $\phi_p > 1$ denotes the gross markup of intermediate goods firms. The parameter $\psi_p \leq 0$ governs the degree of curvature of the intermediate firm's demand curve. If $\psi_p = 0$, $G_Y(\cdot)$ is the standard Dixit and Stiglitz (1977) aggregator. If $\psi_p < 0$, $G_Y(\cdot)$ is the Kimball (1995) aggregator as used in e.g. Smets and Wouters (2007).⁷ Note that $G_Y(1) = 1$, implying constant returns to scale when all intermediate firms produce the same amount.

Final goods producers minimize the cost of producing a given quantity of output Y_t taking as given the price $P_{f,t}$ of each intermediate good $Y_{f,t}$. Final goods producers sell units of final output at a price P_t which they also take as given.

Formally, producers solve: $\max_{Y_t, Y_{f,t}} P_t Y_t - \int_0^1 P_{f,t} Y_{f,t} df$ subject to $\int_0^1 G_Y\left(\frac{Y_{f,t}}{Y_t}\right) df = 1$ where $G_Y(\cdot)$ is given by Eq. (1). The first order conditions can be written as:

$$\frac{Y_{f,t}}{Y_t} = \frac{1}{1+\psi_p} \left(\frac{P_{f,t}}{P_t \vartheta_t^p} \right)^{\varepsilon_p} + \frac{\psi_p}{1+\psi_p} \quad (2)$$

$$1 = \int_0^1 \left(\frac{P_{f,t}}{P_t \vartheta_t^p} \right)^{\frac{\varepsilon_p}{\omega_p}} df \quad (3)$$

$$\vartheta_t^p = 1 + \psi_p - \psi_p \int_0^1 \frac{P_{f,t}}{P_t} df \quad (4)$$

where $\varepsilon_p = \frac{\phi_p(1+\psi_p)}{1-\phi_p}$ and ϑ_t^p denotes the Lagrange multiplier on the aggregator constraint (1). Eq. (2) denotes the demand for goods, Eq. (3) is the aggregate price index and Eq. (4) is the zero profit condition for final goods firms. Note that for $\psi_p = 0$ we obtain the standard Dixit–Stiglitz demand equation and aggregate price index.

Intermediate goods production A continuum of intermediate goods $Y_{f,t}$ for $f \in [0, 1]$ is produced by monopolistically competitive firms, each of which produces a single differentiated good. Each intermediate good producer faces the demand curve derived above.

Production of intermediate good firms is given by $Y_{f,t} = K_f^\alpha N_{f,t}^{1-\alpha}$. Intermediate good firms rent capital and labor services from economy-wide factor markets. Aggregate capital, K , is assumed to be fixed. Even though the aggregate capital stock is fixed, shares of it can be freely allocated across intermediate good firms, i.e. $K \equiv \int K_f df$. Given this, real marginal cost, $MC_{f,t}/P_t$ is identical across firms and equal to $\frac{MC_t}{P_t} \equiv \frac{W_t/P_t}{MPL_t} = \frac{W_t/P_t}{(1-\alpha)(K/N_t)^\alpha}$. We discuss the determination of the aggregate labor-index N_t in Section 3.1.3.

Prices of intermediate goods are determined by Yun (1996) style staggered nominal contracts. In each period, each firm f faces a probability $1 - \xi_p$ of being able to choose its profit maximizing price. Firms solve:

$$\max_{p_{f,t}^{opt}} E_t \sum_{s=0}^{\infty} (\beta \xi_p)^s \varsigma_{t+s} \Lambda_{t+s} (\Pi^s p_{f,t}^{opt} - MC_{t+s}) Y_{f,t+s} \quad (5)$$

where Λ_{t+s} is household marginal utility, ς_{t+s} is a household discount factor shock discussed in Section and demand $Y_{f,t+s}$ is given by Eq. (2).

If a firm does not optimize its price in a given period, it updates its price according to $\tilde{P}_t = \bar{\Pi} P_{t-1}$, where $\bar{\Pi}$ is the steady-state gross inflation rate, i.e. $\bar{\Pi} = 1 + \bar{\pi}$ where $\bar{\pi}$ the is the net steady state price inflation rate.

3.1.2. Intuition

To provide intuition about the implications of the Kimball vs. Dixit–Stiglitz aggregator, we consider the flexible price version of the price setting problem of firms.

The upper left panel in Fig. 1 shows how relative demand in Eq. (2) is affected by the relative price under alternative assumptions about ψ_p , given a gross markup of $\phi_p = 1.1$ and assuming that $\vartheta_t^p = 1$. When $\psi_p = 0$, the demand curve exhibits the standard Dixit–Stiglitz constant demand elasticity, implying a log-linear relationship between relative demand and relative price, i.e. $\log y_f = \varepsilon_p \log p_f$ where $y_f \equiv Y_f/Y$ and $p_f \equiv P_f/P$. With the Kimball specification $\psi_p < 0$ – as in e.g. Smets and Wouters (2007) – a firm instead faces a quasi-kinked demand curve, implying that a drop in its relative price triggers only a small increase in demand. On the other hand, a rise in its relative price generates a larger fall in demand compared to the $\psi_p = 0$ case. Relative to the standard Dixit–Stiglitz aggregator, this introduces more strategic complementarities (or real rigidities) in price setting.

The upper right panel of Fig. 1 depicts the period-by-period profit function $\Phi_f(p_f) = (p_f - mc)y_f(p_f)$ of intermediate good firm f taking into account relative demand (Eq. (2)). More strategic complementarities in the Kimball specification imply that the profit function $\Phi_f(p_f)$ becomes notably more curved when $\psi_p < 0$, especially when the firm's relative price exceeds unity.

⁷ The parameter used in Smets and Wouters (2007) to characterize the curvature of the Kimball aggregator can be mapped to our model using the following formula: $\varepsilon_p = -\frac{\phi_p}{\phi_p - 1} \psi_p$.

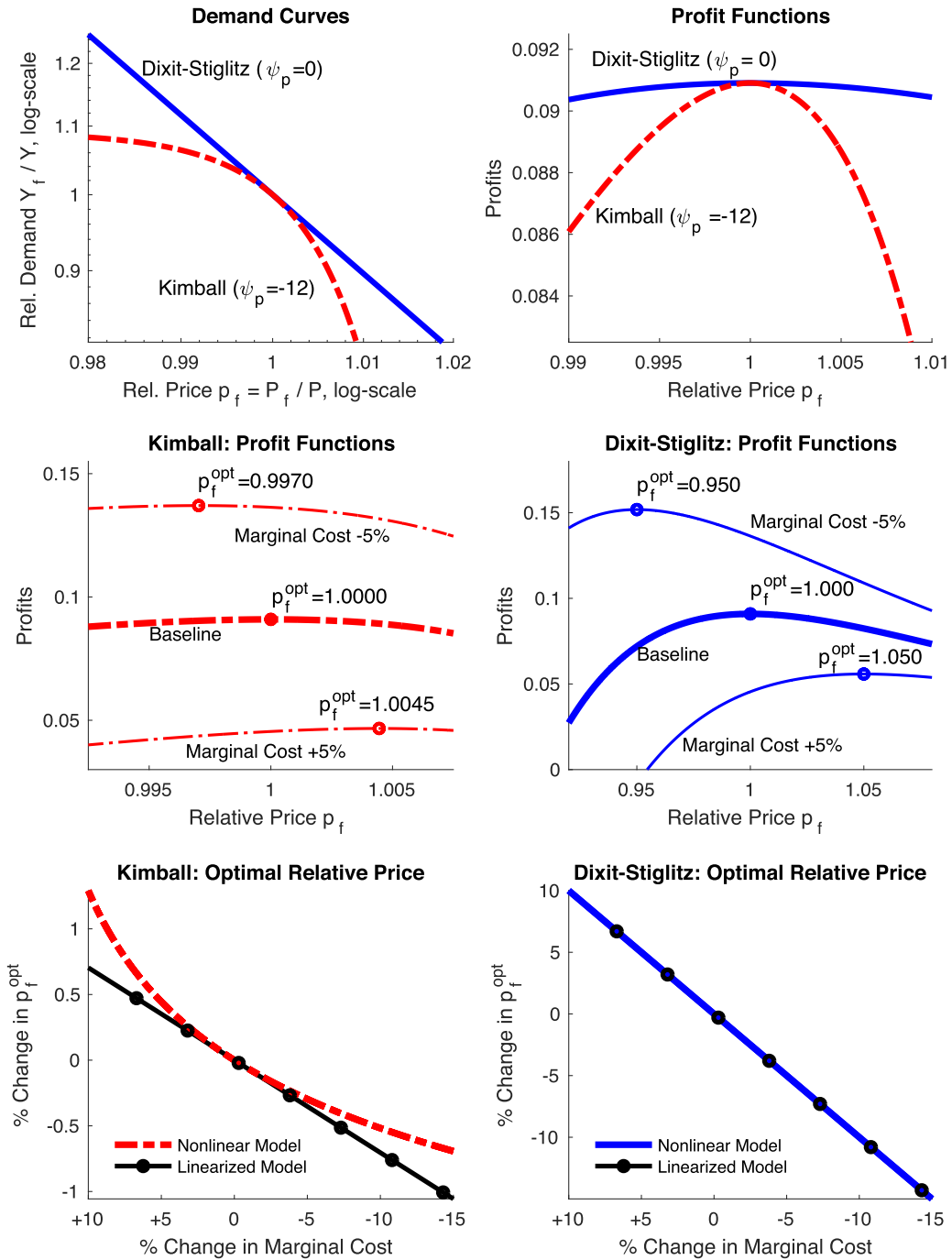


Fig. 1. Kimball vs. Dixit-Stiglitz – demand curves, profits and optimal prices.

The two middle panels in Fig. 1 show how 5 percent positive and negative changes in marginal cost affect firm profits $\Phi_f(p_f)$ and the optimal profit maximizing relative price p_f^{opt} . The left panel shows the effects for the Kimball specification and the right panel shows the effects for the Dixit-Stiglitz specification. The profit function shifts up when marginal cost decline and shifts down when marginal cost rise.

In the Dixit-Stiglitz case ($\psi_p = 0$), the optimal relative price is changed by exactly the same percent change in marginal costs. This result reflects the optimal markup pricing rule implied by maximizing profits subject to the demand curve:

$$\text{Dixit-Stiglitz (nonlinear): } p_f^{opt} = \phi_p mc.$$

Log-linearizing around steady state yields:

$$\text{Dixit–Stiglitz (log-linear): } \frac{p_f^{opt} - \bar{p}_f^{opt}}{\bar{p}_f^{opt}} = \frac{mc - \bar{mc}}{\bar{mc}}$$

where ‘bar’ variables denote steady state variables. Note that the nonlinear and log-linearized optimal pricing rules under Dixit–Stiglitz have the same implications: the optimal relative price changes one to one – in percent terms – with marginal cost.

With the Kimball specification ($\psi_p < 0$), the results can differ sharply from the Dixit–Stiglitz case. Maximizing profits subject to the demand curve yields:

$$\begin{aligned} \text{Kimball (nonlinear): } p_f^{opt} & \left(1 - \frac{\theta_p \psi_p}{1 + \psi_p + \theta_p \psi_p} (p_f^{opt})^{\frac{(1+\theta_p)(1+\psi_p)}{\psi_p}} \right) \\ & = \frac{(1 + \psi_p)(1 + \theta_p)}{1 + \psi_p + \theta_p \psi_p} mc. \end{aligned} \quad (6)$$

where θ_p denotes the net price markup, i.e. $\phi_p = 1 + \theta_p$. The left middle panel in Fig. 1 shows that a negative shock to mc implies a smaller change in p_f^{opt} in absolute value (decline from 1 to 0.997) than a positive shock to mc (increase from 1 to 1.0045) in the nonlinear framework. The asymmetric response of p_f^{opt} to negative and positive shocks to mc in the nonlinear model is apparent from Eq. (6). The equation implies that the asymmetry will be stronger the smaller (i.e. negative) ψ_p and the larger the change in mc .

What is the intuition for the asymmetric response of p_f^{opt} when $\psi_p < 0$? The nonlinearity embedded in the Kimball aggregator and the resulting nonlinear demand curve are key to understand the asymmetric price response. Consider a fall in marginal cost, i.e. a recession. Recall the profit function $\Phi_f(p_f) = (p_f - mc)y_f(p_f)$. Keep in mind that the demand elasticity falls when the relative price of a firm falls. Thus, at the margin, a firm’s ability to increase its demand $y_f(p_f)$ by cutting its price is limited, especially the deeper the recession. Large price cuts result in lower profits because at the margin, demand $y_f(p_f)$ rises only by little while revenues $p_f y_f(p_f)$ fall substantially. Therefore, firms have little incentive to cut prices a lot. The incentive to cut prices becomes weaker the larger the fall in marginal cost since the lower relative price of a firm reduces the demand elasticity implying less demand can be crowded-in by the firm when cutting its price. All told, firms cut their price by less at the margin the deeper the recession, i.e. an asymmetric price response. By continuity, firms change their prices by more at the margin the larger the boom in the economy.

Log-linearizing Eq. (6) around steady state yields:

$$\text{Kimball (log-linear): } \frac{p_f^{opt} - \bar{p}_f^{opt}}{\bar{p}_f^{opt}} = \frac{1}{1 - (1 + \theta_p)\psi_p} \frac{mc - \bar{mc}}{\bar{mc}}$$

Note that in contrast to Dixit–Stiglitz, the optimal relative price does not move one to one in percent terms with marginal costs under the Kimball specification. With Kimball, firms adjust their prices less than with Dixit–Stiglitz since $\frac{1}{1 - (1 + \theta_p)\psi_p} < 1$ when $\psi_p < 0$.

How different are the nonlinear and log-linearized pricing rules implied by Kimball vs. Dixit–Stiglitz quantitatively? The bottom left panel of Fig. 1 plots the optimal relative price p_f^{opt} as a function of marginal cost in the nonlinear vs. log-linear Kimball specification. The bottom right panel shows the Dixit–Stiglitz case. Several important results emerge. First, the nonlinear Kimball specification generates substantial differences between the nonlinear and log-linearized pricing rules. The nonlinear pricing rule implies a banana- or boomerang type pricing schedule: small price adjustments following deep recessions (declines in mc) and large price adjustments following big booms (increases in mc). Second, the nonlinearity becomes stronger the deeper the recession or the bigger the boom. Hence, linearization can produce large approximation errors. Third, the slope of the linearized optimal pricing schedule under Kimball is noticeably flatter compared to the Dixit–Stiglitz case. Fourth, the nonlinear and log-linear pricing schedules under Dixit–Stiglitz are identical. Fifth and finally, when prices are flexible, Kimball implies a countercyclical markup while Dixit–Stiglitz implies a constant markup.

3.1.3. Households and wage setting

Labor contractors Competitive labor contractors aggregate specialized labor inputs $N_{j,t}$ supplied by households into homogenous labor N_t which is hired by intermediate good producers. Labor contractors maximize profits $\max_{N_{j,t}, N_t} W_t N_t - \int_0^1 W_{j,t} N_{j,t} dj$ where $W_{j,t}$ is the wage paid by the labor contractor to households for supplying type j labor. W_t denotes the wage paid to the labor contractor for homogenous labor. Profit maximization is subject to $\int G_N \left(\frac{N_{j,t}}{N_t} \right) dj = 1$ where

$$G_N \left(\frac{N_{j,t}}{N_t} \right) = \frac{\omega_w}{1 + \psi_w} \left((1 + \psi_w) \frac{N_{j,t}}{N_t} - \psi_w \right)^{\frac{1}{\omega_w}} + 1 - \frac{\omega_w}{1 + \psi_w}$$

is the Kimball aggregator specification as used in [Dotsey and King \(2005\)](#) or [Levin et al. \(2007\)](#) adapted for the labor market. Note that $\omega_w = \frac{(1+\psi_w)\phi_w}{1+\phi_w\psi_w}$ where $\phi_w \geq 1$ denotes the gross wage markup and $\psi_w \leq 0$ is the Kimball parameter that controls the degree of complementarities in wage-setting.

Let ϑ_t^w denote the multiplier on the labor contractor's constraint. Optimization results in the following optimality conditions:

$$\frac{N_{j,t}}{N_t} = \frac{1}{1 + \psi_w} \left(\frac{W_{j,t}}{W_t \vartheta_t^w} \right)^{\varepsilon_w} + \frac{\psi_w}{1 + \psi_w} \quad (7)$$

$$1 = \int_0^1 \left(\frac{W_{j,t}}{W_t \vartheta_t^w} \right)^{\frac{\varepsilon_w}{\omega_w}} dj \quad (8)$$

$$\vartheta_t^w = 1 + \psi_w - \psi_w \int_0^1 \frac{W_{j,t}}{W_t} dj \quad (9)$$

where $\varepsilon_w = \frac{\phi_w(1+\psi_w)}{1-\phi_w}$. [Eq. \(7\)](#) denotes the demand for labor, [Eq. \(8\)](#) is the aggregate wage index and [Eq. \(9\)](#) is the zero profit condition for labor contractors. Note that for $\psi_w = 0$ we get the standard Dixit–Stiglitz demand curve and aggregate wage index.

Households There is a continuum of households $j \in [0, 1]$ in the economy. Each household supplies a specialized type of labor service j to the labor market, for which it is a monopolistic supplier. The j th household maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \varsigma_t \left(\ln C_{j,t} - \omega \frac{N_{j,t}^{1+\chi}}{1+\chi} \right) \quad (10)$$

subject to

$$P_t C_{j,t} + B_{j,t} = W_{j,t} N_{j,t} + R_t^k K_j + (1 + i_{t-1}) B_{j,t-1} - T_{j,t} + \Gamma_{j,t} + A_{j,t}$$

where the choice variables of the j th household are consumption $C_{j,t}$ and risk-free government debt B_t . The j th household also chooses the wage $W_{j,t}$ subject to Calvo sticky prices as in EHL. The household understands that when choosing $W_{j,t}$ it must supply the amount of labor $N_{j,t}$ demanded by a labor contractor according to [Eq. \(7\)](#).

The variable ς_t is an exogenous shock to the discount factor $0 < \beta < 1$. We assume that $\delta_t = \frac{\varsigma_t}{\varsigma_{t-1}}$ is exogenous with $\delta = 1$ in steady state. P_t denotes the aggregate price level. i_t denotes the net nominal interest rate on bonds purchased in period $t-1$ which pay off in period t . R_t^k is the rental rate of the fixed aggregate capital stock that the households rents to goods producing firms. $T_{j,t}$ are lump-sum taxes net of transfers and $\Gamma_{j,t}$ denotes the share of profits that the household receives. $A_{j,t}$ denotes payments and receipts associated with insurance in the presence of wage stickiness.⁸ $\omega > 0$ and $\chi \geq 0$ are parameters.

Utility maximization for consumption and government bond holdings yields the standard aggregate consumption Euler equation:

$$1 = \beta E_t \left(\delta_{t+1} \frac{1 + i_t}{1 + \pi_{t+1}} \frac{C_t}{C_{t+1}} \right) \quad (11)$$

where $1 + \pi_{t+1} = P_{t+1}/P_t$.

Wage setting Households face a standard monopoly (labor union) problem of selecting $W_{j,t}$ to maximize utility [\(10\)](#) subject to the demand for labor [\(7\)](#). Following EHL, we assume that households experience Calvo-style frictions in its choice of $W_{j,t}$. In particular, with probability $1 - \xi_w$ the j th household has the opportunity to choose its optimal wage $W_{j,t}^{opt}$. With the complementary probability, the household must set its wage rate according to $\tilde{W}_{j,t} = \tilde{\Pi}_w W_{j,t-1}$ where $\tilde{\Pi}_w$ denotes the steady state gross rate of wage inflation. Households choose the optimal wage by maximizing

$$\max_{W_{j,t}^{opt}} E_t \sum_{s=0}^{\infty} (\beta \xi_w)^s \varsigma_{t+s} \left(\Lambda_{t+s} W_{j,t}^{opt} \Pi_w^s N_{j,t+s} - \omega \frac{N_{j,t+s}^{1+\chi}}{1+\chi} \right)$$

subject to labor demand given by [Eq. \(7\)](#).

⁸ In principle, the presence of wage setting frictions implies that households have idiosyncratic levels of wealth and, hence, consumption. However, we follow EHL in supposing that each household has access to perfect consumption insurance. Because of the additive separability of the household utility function, perfect consumption insurance implies equal consumption across households. Note that even though consumption is equal across households, aggregate consumption responds to shocks and is hence not constant over time.

3.1.4. Monetary and fiscal policy

We assume that the central bank sets the nominal interest rate following a Taylor-type policy rule that is subject to the zero lower bound:

$$1 + i_t = \max \left\{ 1, (1 + i) \left(\frac{1 + \pi_t}{1 + \bar{\pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{Y_t^{pot}} \right)^{\gamma_x} \right\} \quad (12)$$

where Y_t^{pot} denotes the level of output that would prevail if prices and wages were flexible.

In terms of fiscal policy, we assume that the government balances its budget each period using lump-sum taxes.

3.1.5. Aggregate resource constraint

It is straightforward to show that the aggregate resource constraint is given by

$$C_t = Y_t = (p_t^*)^{-1} (w_t^*)^{-(1-\alpha)} K^\alpha l_t^{1-\alpha}$$

where

$$p_t^* \equiv \frac{1}{1+\psi_p} \int_0^1 \left(\frac{p_{f,t}}{p_t} \right)^{-\varepsilon_p} df + \frac{\psi_p}{1+\psi_p}$$

$$w_t^* \equiv \frac{1}{1+\psi_w} \int_0^1 \left(\frac{w_{j,t}}{w_t} \right)^{-\varepsilon_w} dj + \frac{\psi_w}{1+\psi_w}.$$

Aggregate capital K is given by $K = \int K_f df = \int K_j dj$. Aggregate hours per capita l_t supplied by households is given by $l_t = \int N_{j,t} dj = w_t^* N_t$ where $N_t = \int N_{f,t} df$. The variables $p_t^* \geq 1$ and $w_t^* \geq 1$ denote the Yun (1996) aggregate price and wage dispersion terms. Both price- and wage-dispersion, ceteris paribus, will lower aggregate output in the economy. The Technical Appendix provides recursive formulations for the sticky price and wage distortion terms p_t^* and w_t^* . Note that p_t^* and w_t^* vanish when the model is linearized.

3.2. Parameterization

The model is parameterized using pre-Great Recession empirical evidence. Doing so allows us to assess how well the model accounts for the observed patterns of output and inflation during the Great Recession using pre-Great Recession parameters.

Time is taken to be quarters. The discount factor is set to $\beta = 0.9975$ and the steady state net inflation rate $\pi = 0.005$ implying a steady state nominal interest rate of $i = 0.0075$ (i.e., three percent at an annualized rate). The capital share parameter is set to $\alpha = 0.3$ and the disutility of labor parameter $\chi = 0$. As a compromise between the low estimate of the gross price markup ϕ_p in Altig et al. (2011) and the higher estimated value by Smets and Wouters (2007), we set $\phi_p = 1.1$.

Consistent with microeconomic evidence in e.g. Nakamura and Steinsson (2008) we set $\xi_p = 0.66$, i.e. prices change on average every three quarters. To pin down the Kimball parameter ψ_p consider the log-linearized New Keynesian Phillips curve in our model: $\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \kappa_p \widehat{mc}_t$, where $\widehat{mc}_t$ denotes the log-deviation of real marginal cost from its steady state. $\hat{\pi}_t$ denotes the log-deviation of gross inflation from its steady state. The parameter κ_p denotes the slope of the Phillips curve and is given by $\kappa_p \equiv \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p} \frac{1}{1-\phi_p\psi_p}$.

The macroeconomic evidence suggests that the sensitivity of aggregate inflation to variations in real marginal cost is very low, see e.g. Altig et al. (2011). To capture this, the Kimball parameter is set to $\psi_p = -12.2$ so that the slope of the Phillips curve is $\kappa_p = 0.012$ given the values for β , ξ_p and ϕ_p discussed above.⁹

For the parameters pertaining to the nominal wage setting frictions set the wage markup equal to the price markup, i.e. $\phi_w = 1.1$. Further, the Calvo wage stickiness parameter is set to $\xi_w = 0.75$. Finally, the wage Kimball parameter is set to $\psi_w = -6$ reflecting a somewhat steeper slope of the wage Phillips curve compared to the price Phillips curve in DSGE models (see e.g. Smets and Wouters, 2007). The Taylor (1993) rule is set to the standard parameters $\gamma_\pi = 1.5$ and $\gamma_x = 0.125$.

To facilitate comparison between the nonlinear and linearized model, we specify an AR(1) process for the discount factor so that there is no loss in precision due to an approximation:

$$\delta_t - \delta = \rho_\delta (\delta_{t-1} - \delta) + \sigma_\delta \varepsilon_{\delta,t} \quad (13)$$

where $\delta = 1$. Our baseline parameterization adopts a persistence coefficient $\rho_\delta = 0.95$ capturing the persistent drop in output during the Great Recession.

⁹ The median estimates of the Phillips curve slope in pre-Great Recession empirical studies by e.g. Adolfson et al. (2005); Altig et al. (2011); Galí et al. (2001); Galí and Mark (1999); Lindé (2005), and Smets and Wouters (2003, 2007) are in the range of 0.009–0.014.

3.3. Solving the model

Our baseline results for the linearized and nonlinear model are based on the Fair and Taylor (1983) solution algorithm. This solution algorithm is also known as a two-point boundary value solution algorithm or time-stacking algorithm. The Fair–Taylor solution algorithm imposes certainty equivalence on the nonlinear model, just as the linearized model solution does by definition. In other words, the Fair–Taylor solution algorithm allows us to trace out the implications of not linearizing the equilibrium equations which is exactly our objective. All relevant information for solving the nonlinear and linearized model is captured by the current state of the economy, including the contemporaneous realization of the exogenous discount factor shock.

An alternative approach to solve the model would have been to compute solutions where uncertainty about future shock realizations matters for the dynamics of the economy following e.g. Adam and Billi (2006, 2007); Aruoba et al. (2018); Fernández-Villaverde et al. (2015b); Gust et al. (2017) and Nakata (2017). These authors have shown that allowing for future shock uncertainty can have potentially important implications for equilibrium dynamics. Importantly, none of these authors have considered a model with Kimball aggregation. Lindé and Trabandt (2018) solve a simplified version of our model with sticky prices and Kimball aggregation under shock uncertainty using global methods, and show that the effects of future shock uncertainty on the global solution of the nonlinear model are quantitatively negligible lending support for using the Fair–Taylor solution method for our baseline results.¹⁰

As a practical matter, the equilibrium equations of the nonlinear and linearized model are fed into Dynare. Dynare is a pre-processor and a collection of MATLAB routines which can solve nonlinear and linearized dynamic models with forward looking variables. The details about the implementation of the algorithm used can be found in Juillard (1996). We use the perfect foresight/deterministic simulation algorithm implemented in Dynare using the ‘simul’ command.¹¹ The algorithm can also easily handle the ZLB constraint: one just writes the Taylor rule including the max operator in the model equations, and the solution algorithm reliably calculates the model solution in fractions of a second. Thus, apart from obtaining intuition about the various mechanisms embedded in dynamic models, there is no need anymore to linearize dynamic models to solve, simulate and work with them.

4. Inflation-output dynamics in the stylized model

This section reports our key results for the linearized and nonlinear solution of the model outlined in the previous section.

4.1. Phillips curves

To understand the unconditional differences in the dynamics implied by the linearized and nonlinear solutions, we now undertake stochastic simulations of the model for shocks to the stochastic discount factor δ_t . The linearized and nonlinear models are solved and simulated for a long sample of 50,000 periods contingent on exactly the same sequence of shocks $\{\varepsilon_{\delta,t}\}_{t=1}^{50,000}$ in Eq. (13). However, we use somewhat different standard deviations for the linearized model ($\sigma_\delta = 0.0014$) and the nonlinear model ($\sigma_\delta = 0.0017$), to ensure that the probability of hitting the ZLB is 10 percent in both model solutions.

The simulated time series, shown in Figure A.6 in the Technical Appendix, show noticeable differences between the linearized and nonlinear model for the behavior of price and wage inflation. The simulated data from the linearized model shows several episodes with substantial deflation, whereas the nonlinear model does not feature any spells with deflation in prices and wages. There are several episodes when price and wage inflation is persistently low, but there are no spells with deflationary outcomes because the Kimball aggregator implies that firms (unions) are reluctant to change prices (wages) much when relative demand is low. On the other hand, in periods when relative demand is high, firms (unions) are more willing to change their prices. As a result, the nonlinear model produces episodes with more elevated price and wage inflation than the linearized solution in which households and firms are – except for ZLB periods – equally sensitive to changes in desired price and wage markups in recessions and booms.

While the simulation results are instructive to understand many features of the nonlinear model, it is not straightforward to relate the behavior of price and wage inflation to the state of the business cycle. The relationship between actual price/wage inflation and some measure of resource utilization is traditionally referred to as the Phillips (1958) drew this original downward-sloping relationship between the rate of wage inflation and the unemployment rate. Subsequently, influential work by Clarida et al. (1999); Roberts (1995) and Blanchard (2000) has extended Phillips’ approach to the relationship between price inflation and the output gap. Thus, we use the simulated data to produce bivariate scatter plots between price (and wage) inflation on the y-axis and the negative of the output gap on the x-axis. The negative output gap is the differ-

¹⁰ The introduction of wage stickiness and Kimball aggregation in the labor market in the present paper (in addition to price stickiness and Kimball aggregation in the goods market as in Lindé and Trabandt (2018) should temper the effect of shock uncertainty in the nonlinear model even further. To the extent that allowing for shock uncertainty impacts notably the linearized solution, the differences between the linearized and nonlinear solutions we report in this paper are conservative: they would be even larger if we had allowed for shock uncertainty.

¹¹ The solution algorithm implemented in Dynare’s simul command is the method developed in Fair and Taylor (1983).

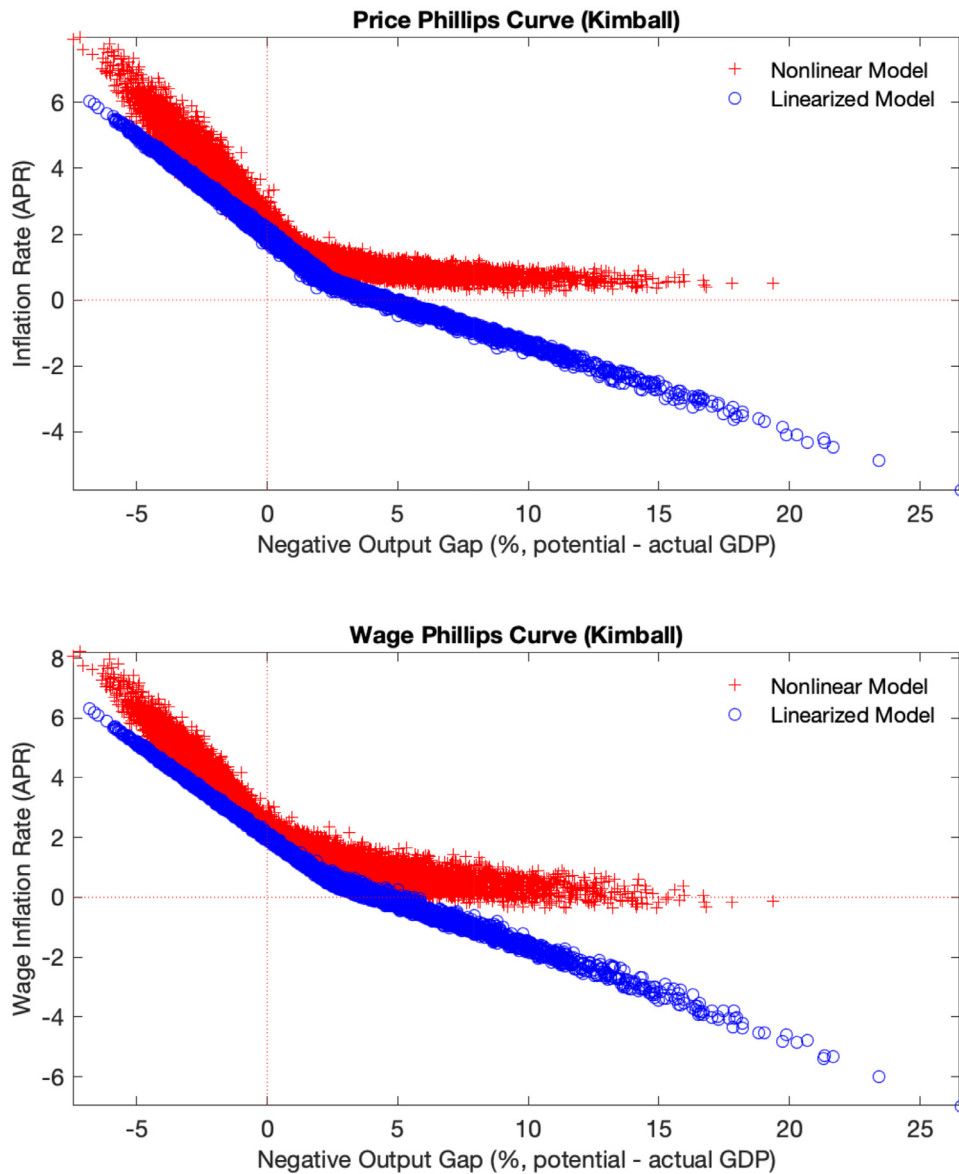


Fig. 2. Price and wage Phillips curves.

ence between potential output and actual output. By using the negative of the output gap, we derive a downward-sloping relationship as in [Phillips \(1958\)](#).

[Fig. 2](#) shows the results, with the price- and wage-inflation Phillips curves in the upper- and lower panel, respectively. Given that the x-axis plots the negative output gap, i.e. potential minus actual real GDP, a positive number on the x-axis corresponds to recessions and a negative number on the x-axis corresponds to booms. In the linearized model (blue circles), we see that the relationship between wages and prices and the negative output gap is characterized by a constant negative slope coefficient around the steady state rate of inflation of two percent.¹² However, when the economy hits the ZLB, which tends to happen when the *negative of the output gap* is around or above 2.5 percent, then the slopes of the linear price and wage Phillips curves flatten somewhat, implying that the output gap is more strongly affected by discount factor shocks than price and wage inflation. At the ZLB, the nominal interest rate is constant and further declines in inflation elevate the real interest rate gap which amplifies the decline in economic activity even more – hence a somewhat flatter slope. Even so, the implied flattening of the linearized Phillips curves due to the ZLB is not nearly enough to resolve the missing inflation

¹² In the EHL model, we do not allow for permanent productivity growth which would raise nominal wage inflation above price inflation in the steady state. We relax this simplifying assumption in the estimated model in Section 5.

puzzle. The linearized model erroneously predicts a much larger fall in inflation by extrapolating local decision rules around the steady state to deep recessions far away from the steady state.

Section A.6 in the Technical Appendix provides an in-depth analysis and intuition why the joint output-inflation dynamics can differ so much in response to large adverse demand shocks.

By contrast, and consistent with the intuition developed in [Section 3.1.2](#), the nonlinear model with the Kimball aggregator (red crosses) implies nonlinear price and wage Phillips curves, i.e. nonlinear relationships between price and wage inflation and the negative output gap. The slopes of our price and wage Phillips curves are notably flatter in recessions than in booms, i.e. our nonlinear Phillips curves have a banana- or boomerang-shape as in the seminal paper by [Phillips \(1958\)](#). Consequently and consistent with the data, inflation falls relatively little in deep recessions in our nonlinear model. Through the lens of our model, the missing deflation puzzle arises when working with the linearized version of the underlying nonlinear model.

This finding is very interesting as it – in addition to explaining why inflation fell so little during the Great Recession – offers a possible explanation why prices and wages have risen so little when advanced economies recovered from the Great Recession. As evidenced by [Fig. 2](#), our nonlinear model implies that price and wage inflation remain subdued until the level of economic activity has recovered sufficiently relative to its potential for our demand shock. Put differently, even though economic growth may resume after a deep recession, price and wage inflation will only increase noticeably when economic slack has subsided sufficiently. Therefore, our nonlinear model does not only resolve the missing deflation puzzle during the Great Recession, it also offers an explanation for the so-called missing inflation puzzle observed in the aftermath of the Great Recession (see [IMF, 2016; IMF, 2017](#)). According to our model, subdued price and wage inflationary pressures as observed during the recovery from the Great Recession are indicative of the presence of substantial economic slack. While the linearized model suggests that price and wage inflation should rise sharply during a recovery from a deep recession, the nonlinear model suggests otherwise: price and wage inflation will not rise until economic slack has narrowed sufficiently from its elevated recession level.

Moreover, the nonlinear model also features stronger responses of price and wage inflation in booms than the linearized model, lending support for inflation scares in booms (see e.g. [Goodfriend, 1993](#)). Another difference between the linearized and nonlinear solutions is that the output gap is more volatile in the linearized solution, mainly due to strong propagation of the ZLB constraint. Finally, it is imperative to understand that the relationships in [Fig. 2](#) are contingent on the assumption that the discount factor shock is the single driver of business cycles. No other shocks are assumed to affect the economy. This is why we obtain a tight negative relationship between price and wage inflation and economic slack. We now turn to assess these findings in a workhorse medium-sized New Keynesian model where several shocks affect the economy simultaneously.

5. Inflation dynamics during the great recession

The benchmark EHL model studied so far is useful to understand how nonlinearities in real rigidities in price- and wage-setting affect inflation dynamics in deep recessions. Specifically, we used the EHL model to demonstrate some of the benefits of taking nonlinearities into account as opposed to the traditional approach which entails log-linearizing the key model equations apart from the monetary policy rule.

However, this analysis was done in a stylized model with one shock and without allowing for e.g. endogenous capital accumulation. In this section we move on to a substantive analysis with the aim of examining the importance real rigidities in a nonlinear setting in a more quantitatively realistic model environment. Specifically, we use the workhorse [Smets and Wouters \(2007\)](#) model of the US economy which is a multishock version of the seminal model of [Christiano et al. \(2005\)](#).

In the following, we first give a brief overview of the model and describe how the filtering of shocks and likelihood computation in the linearized and nonlinear solutions of the model are done. Next, historical decompositions of inflation without the markup shocks are shown to highlight the linearized models' dependency on positive markup shocks to account for the missing deflation puzzle whereas the nonlinear solution can avoid deflation without such shocks. After this, the log-likelihood of the linearized and nonlinear model during the Great Recession is discussed. Finally, the ability of the nonlinear model to capture the unconditional skewness of price and wage inflation is examined.

5.1. Model overview and filtering of shocks

As in the EHL model described in [Section 3](#), the workhorse model of [Smets and Wouters \(2007\)](#) features monopolistic competition in the goods and labour markets and nominal frictions in the form of sticky prices and wages. However, it allows non-optimizing firms (households) to index prices (wages) to a composite of steady-state and lagged inflation. Households can also save in physical capital, with a one-period time to build before new investments turns into productive capital. In addition to the EHL model, the Smets and Wouters model also features several real rigidities in the form of habit formation in consumption, investment adjustment costs, variable capital utilization, and fixed costs in production. The model dynamics are driven by seven structural shocks. Monetary policy shocks follow an AR(1) process and two additional inefficient markup shocks in wage and price setting follow an ARMA(1,1) process. Four efficient shocks (total factor productivity, risk premium, investment-specific technology, and government spending shocks) follow AR(1) processes. The exact specification of the nonlinear model and its linearized representation is described in detail in [Appendix B](#).

The model is estimated on seven key macroeconomic quarterly U.S. time series as observable variables: the log differences of real per capita GDP, consumption and investment; log-differences of compensation per hour and the GDP deflator; log deviations of hours per capita from its average as well as the Federal Funds rate. Further details about the data and the measurement equations linking the model variables to their data counterparts are provided in the Appendix.

Smets and Wouters (2007) use full information Bayesian techniques to estimate the model. Bayesian inference starts out from a prior distribution that describes the available information prior to observing the data used in the estimation. The observed data is subsequently used to update the prior, via Bayes' theorem, to a posterior distribution of the model's parameters which can be summarized in the usual measures of location (e.g. mode or mean) and spread (e.g. standard deviation and probability intervals).¹³

To begin with, we adopt the SW posterior mode estimates (and their calibrated parameters). The parameters are tabulated in the appendix. Conditional on these parameters, we filter the shocks in the linearized and nonlinear solutions with the inversion filter described in Fair and Taylor (1983) from 1965Q1 to 2014Q2.¹⁴ Since we have the same number of observables as shocks, the inversion filter provides us with a unique sequence of filtered innovations $\{\varepsilon_{t|t}\}_{t=1}^T$ that maximizes the likelihood for a given parameterization of the linearized and nonlinear model.¹⁵ Given the large number of state variables and shocks in the SW model, it takes about 45 min on an 8-core CPU workstation to compute the filtered shocks in the nonlinear model. This renders estimation of the parameters of the nonlinear model infeasible at this point in time.

Conditional on period t information, let $X_{t|t}$ denote all variables in the model, except for the innovations to the shock processes $\varepsilon_{t|t}$. The solution of the model is a nonlinear function f such that

$$X_{t|t} = f(X_{t-1|t-1}, \varepsilon_{t|t}), \quad (14)$$

and let y_t be a vector of observables and S a selection matrix that matches the observed variables to their model counterparts. Then, in each period t the filtered structural innovations $\varepsilon_{t|t}$ are obtained, given the data y_t and state of the model $X_{t-1|t-1}$, as the solution to the nonlinear system

$$y_t = Sf(X_{t-1|t-1}, \varepsilon_{t|t}). \quad (15)$$

The filter is initialized by running the Kalman smoother on the linearized solution and using the implied Kalman smoothed estimates as initial values for the endogenous variables in the nonlinear filter. However, the nonlinear model contains some additional endogenous state variables (i.e. price and wage dispersion terms), and these endogenous variables are initialized together with the shock innovations $\varepsilon_{t|t}$ so that the nonlinear model matches the observed variables in y_t .

Importantly, we impose the Effective Lower Bound (ELB) for the federal funds rate when filtering the shocks in the linearized and nonlinear models. Since the quarterly average for the periods 2009Q1–2014Q2 is not exactly zero and fluctuates a couple of basis points, we adjust the federal funds rate to exactly 10 (annualized) basis points for those quarters. Accordingly, the monetary policy rule in the model is constrained by an ELB of 10 (annualized) basis points. One important caveat of using this filtering approach is that, since monetary policy shocks become inconsequential in the model when the ELB is binding, it is not possible to filter these shocks during the ELB period. Following Guerrieri and Iacoviello (2017) these shocks are assumed to be zero when the ELB is binding, except when the model-implied interest rate is predicted to be above the ELB and a monetary policy shock is necessary to obtain an exact match between the observed data and the model counterparts. Finally, note that the Fair and Taylor (1983) solution algorithm we adopt imposes certainty equivalence on the nonlinear model, just as the linearized model solution does by definition.

5.2. Role of nonlinearities during the great recession

We now use the filtered shocks to assess the linearized and nonlinear solutions' dependence on markup shocks to explain the missing deflation puzzle. To parse out the role of markup shocks to explain inflation dynamics during the Great Recession and its aftermath, we simulate counterfactual paths for the linearized and nonlinear solution when setting both price and wage markup innovations to nil from 2008Q4 onwards (all innovations are included before that).

Fig. 3 depicts the findings of this exercise. Panel A shows the results when using the SW parameters. The black line shows actual inflation, the blue dashed line the counterfactual in the baseline linearized model, and the red dotted line the nonlinear counterfactual.¹⁶ As expected given the findings in Fratto and Uhlig (2018), the linearized solution without the markup shocks generates a prolonged period with negative inflation rates, i.e. a deflation puzzle. The nonlinear model generates a somewhat less pronounced but still sizable deflationary episode, with a faster recovery in inflation.

The more limited role of nonlinearities in the SW model relative to the EHL model studied earlier is not surprising given that the Kimball price setting parameter is notably lower in the SW calibration. We will next show that a simple tweak of

¹³ We refer the reader to Smets and Wouters (2003, 2007) for a more detailed description of the estimation procedure.

¹⁴ The SW sample is 1965Q1 through 2004Q4.

¹⁵ A potential drawback of using the inversion filter is that it does not allow to compute the two-sided innovation (i.e. $\varepsilon_{t|T}$), and hence relies on the one-sided filtered shocks $\varepsilon_{t|t}$ in the analysis. That said, in the linearized solution, the one- and two-sided shocks (the latter being based on a standard Kalman filter) are very similar, suggesting that using the one-sided shocks has a limited impact on the findings.

¹⁶ In this decomposition we allow the policy rate in the linearized model to reach zero instead of the 10 basis points ELB mentioned above in order to avoid a deflationary spiral that ensues without this extra accommodative policy space.

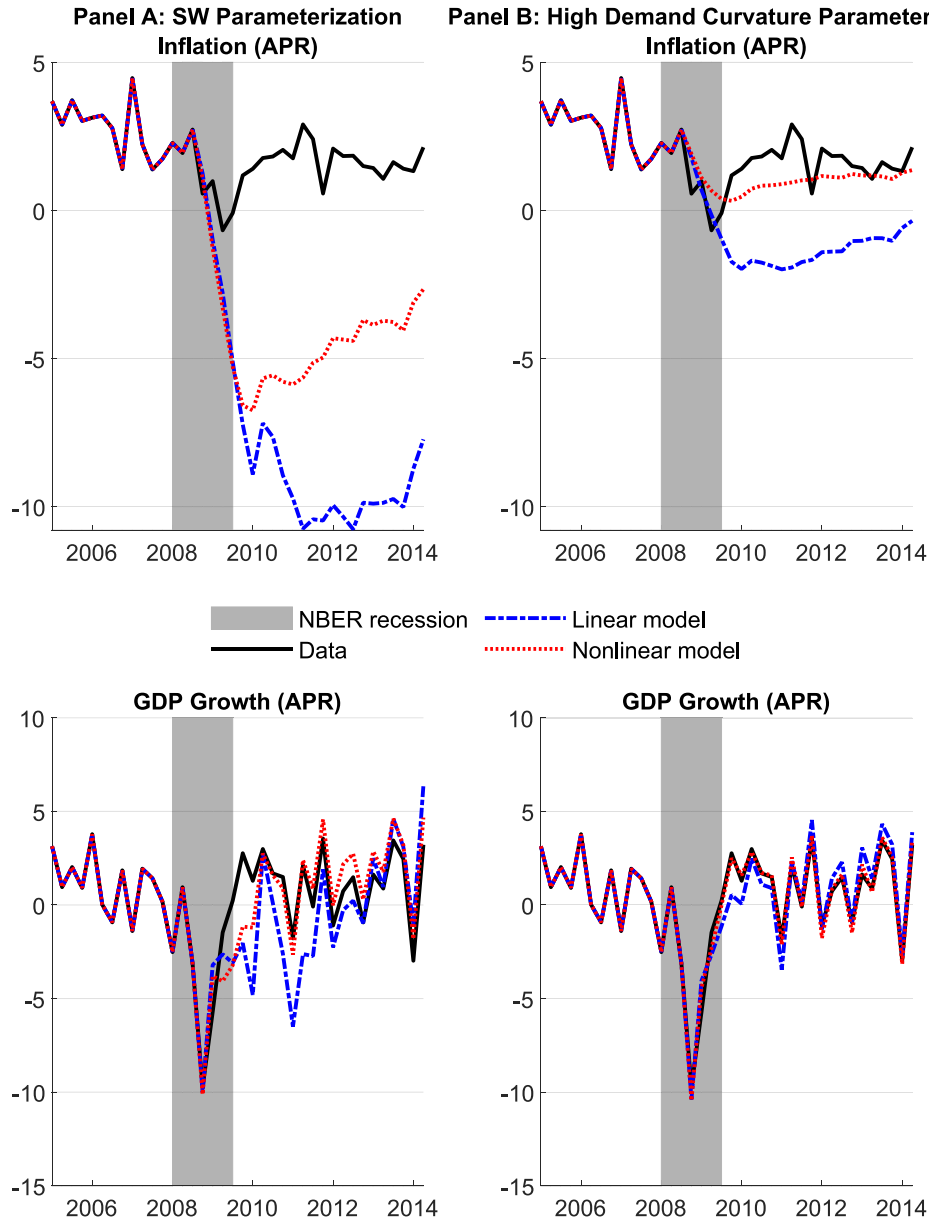


Fig. 3. Inflation and output dynamics without markup shocks.

the SW prior – observationally equivalent from the point of view of the New Keynesian Phillips Curve – allows us to address the missing deflation puzzle in the nonlinear SW model.

To motivate our alternative calibration, notice that the linearized Phillips curve in the SW model is given by

$$\begin{aligned}\hat{\pi}_t - \iota_p \hat{\pi}_{t-1} &= \beta (E_t \hat{\pi}_{t+1} - \iota_p \hat{\pi}_t) + \kappa \widehat{mc}_t + \hat{\varepsilon}_t^p, \\ \kappa &= \frac{(1 - \xi_p \beta)(1 - \xi_p)}{\xi_p(1 + (\phi_p - 1)\epsilon_p)}\end{aligned}\quad (16)$$

where $1 - \xi_p$ is the probability of each firm being able to re-optimize the price each period, ϵ_p is the curvature of the SW aggregator function, and the price markup shock $\hat{\varepsilon}_t^p$ has been re-scaled to enter with a unit coefficient.¹⁷ SW adopted the prior $\phi_p \sim N(1.25, 0.125)$ and obtain a posterior mode $\phi_p = 1.61$, i.e. a markup of 61%. SW also estimate $\xi_p = 0.65$ and calibrate $\epsilon_p = 10$. So, since $\beta = 0.998$ the posterior mode for the slope of the Phillips curve is $\kappa = .026$.

¹⁷ Notice that the relationship between the SW Kimball parameter ϵ_p and the Kimball parameter ψ_p in Eq. (1) is given by $\epsilon_p = \frac{\phi_p}{1 - \phi_p} \psi_p$.

Table 1

Markups, Kimball elasticities, Phillips curve slopes and GFC inflation trough.

Gross markup ϕ_p	Kimball elasticity ϵ_p	Phillips curve slope κ	GFC inflation trough (APR)
1.10	94.3	0.016	0.37
1.15	84.2	0.012	0.40
1.20	77.3	0.010	0.38
1.25	72.2	0.009	0.36
1.34 ^a	64.5	0.007	0.32
1.40 ^b	50.1	0.008	-0.19
1.61 ^c	10.0	0.026	-6.75

Note: The table shows results when re-estimating ϵ_p and other SW model parameters for different calibrated values of ϕ_p given $\xi_p = 0.667$. ^a $\phi_p = 1.34$ is the posterior with our alternative prior. ^b 1.40 is the maximum estimated markup in NK DSGE papers not based on the SW model. ^c $\phi_p = 1.61$ is the posterior with the baseline SW prior given $\epsilon_p = 10$.

A large body of work that calibrates or estimates the markup in New Keynesian models which differ from the Smets and Wouters model suggests a markup that is notably lower than 61 percent.¹⁸ This fact motivates us to consider the effects of using an alternative prior $\phi_p \sim N(1.2, 0.05)$ which is tilted somewhat relative to SW when re-estimating the model.¹⁹ Further, we calibrate $\xi_p = 0.667$ in line with micro evidence discussed in Section 3 (which is close to the SW posterior mode of 0.65) and estimate the Kimball curvature parameter using the prior $\epsilon_p \sim N(75, 25)$. This results in a posterior mode where $\epsilon_p = 64.5$ and $\phi_p = 1.34$. Overall, the re-estimated model implies $\kappa = 0.0072$ in Eq. (16) which is somewhat smaller than SW's estimate of 0.026.²⁰ Note that the re-estimated model is associated with an improvement of the marginal likelihood compared to the original SW parameterization with $\epsilon_p = 10$ by roughly 5 log points. So conditional on a prior which makes the posterior for the markup not exceed calibrated or estimated values in the New Keynesian literature, our posterior with a higher Kimball elasticity improves the marginal likelihood in the SW model.

Panel B in Fig. 3 reports the results for this alternative calibration of the model. Comparing the paths with the SW parameterization in Panel A, we see that inflation falls considerably less in both the linearized and nonlinear models. In particular, the nonlinear solution avoids deflation altogether. Hence, it addresses the missing deflation puzzle, whereas the linearized model is still associated with a sustained deflation episode. The lower panels show that the high curvature parameterization also tracks GDP much more closely than the SW benchmark during and after the Great Recession.

To understand the interaction between ϕ_p and ϵ_p in addressing the missing deflation puzzle, Table 1 reports the results when the gross markup ϕ_p is fixed at different values commonly used in the DSGE literature, and the Kimball parameter ϵ_p is re-estimated along with the other parameters in the linearized model. Throughout, the price contract parameter is set to $\xi_p = 0.667$. The table highlights the inverse relationship between ϕ_p and ϵ_p , and that lower values for ϕ_p than our posterior 1.34 also imply that the nonlinear model resolves the missing deflation puzzle. Since 1.4 is the upper bound for calibrated and estimated values for the surveyed papers in the NK DSGE literature we include this value as well along with the SW posterior of $\phi_p = 1.61$ when $\epsilon_p = 10$. In line with Panel A in Fig. 3 we obtain substantial deflation at the trough with the SW parameterization. By contrast, gross markups below 1.40 imply that the nonlinear model resolves the missing deflation puzzle.

Complementing the findings in Fig. 3 and Table 1, Fig. 4 presents two additional model parameterizations that shed light on the role of the steady state markup and Kimball curvature for inflation and output dynamics in the nonlinear model. The dashed yellow line shows the same experiment when replacing the Kimball aggregator with the Dixit–Stiglitz aggregator and adjusting the Calvo parameter such that slope of the linearized Phillips curve (κ in Eq. (16)) remains unchanged (which implies setting $\xi_p = 0.85$). The dashed purple line shows an alternative estimation of our preferred prior for $\phi_p \sim N(1.2, 0.05)$ when we impose an unchanged SW Phillips curve slope (0.026) with an unchanged Kimball curvature $\epsilon_p = 64.5$. This yields an estimated gross steady state markup of 1.08. The dotted lines repeat the dotted red lines in the upper panels A and B of Fig. 3.

Fig. 4 illustrates the crucial role of nonlinearities in the Phillips curve to explain inflation dynamics during and after the Great Recession. While the two dashed lines and the blue dotted line share the identical linearized Phillips curves, they differ in terms of steady state markup and Kimball curvature. The high markup-low Kimball curvature parameterizations show a notably larger deflation puzzle than the high curvature parameterizations, with the Dixit–Stiglitz (no curvature) case triggering an even deeper deflation compared to the benchmark SW parameterization. In contrast, the purple dashed line

¹⁸ When looking at papers that calibrate (Del Negro et al., 2017; Eggertsson and Singh, 2019; Erceg et al., 2000; Fernández-Villaverde et al., 2015a; Galí, 2015; Gust et al., 2017; Nakata, 2017; Woodford, 2003) or estimate (Adolfson et al., 2007; Altig et al., 2011; Christiano et al., 2016; 2005; 2010; Fernández-Villaverde and Rubio-Ramírez, 2008; Rotemberg and Woodford, 1997) gross markups in models, it turns out that the average and median markup is 17 percent and 15 percent, respectively. In no case the gross markup exceeds 1.4, which governs our choice of tightness for the alternative prior.

¹⁹ We re-estimate the linearized model for the sample 1965Q1–2007Q4 and following SW, 1965Q1–1965Q4 is used as training sample and the likelihood function is evaluated from 1966Q1 and onwards. Note that our estimation approach is consistent with SW, who estimate the linearized model with pre-Great Recession data.

²⁰ Our lower value for κ partly reflects the different prior and partly the extended sample period (recalling that SW sample ends 2004:4). For our sample period ending 2007:4, we obtain $\kappa = .018$ with the SW prior.

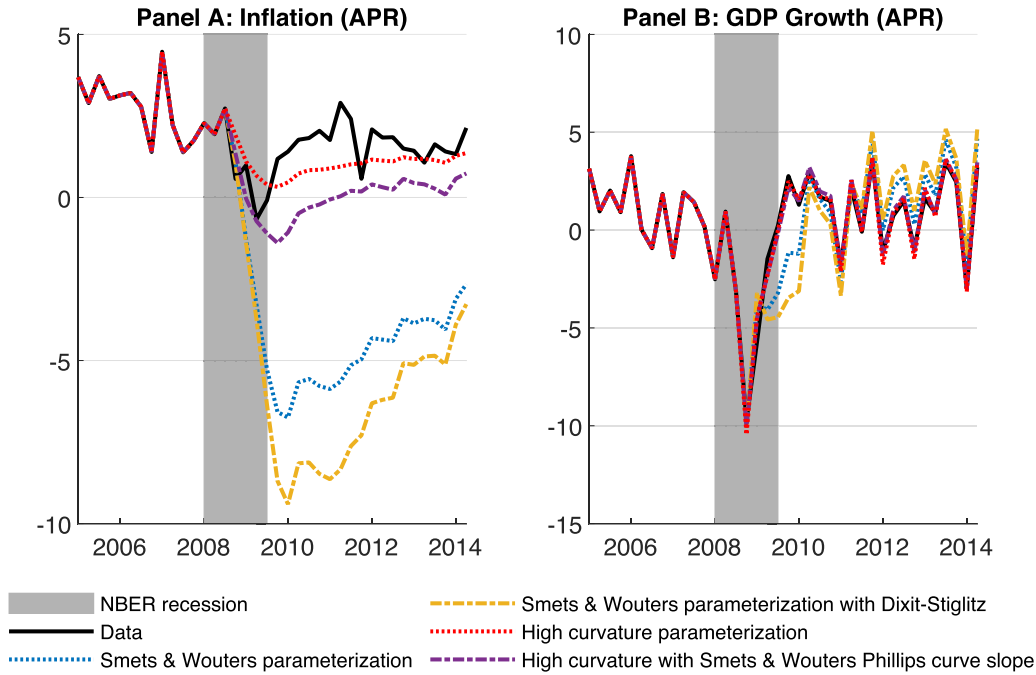


Fig. 4. Decomposition of inflation and output dynamics in the nonlinear model: the role of demand curvature and the Phillips curve slope.

shows that increasing the Kimball curvature while keeping the Phillips curve slope unchanged mutes the deflation puzzle notably. We also see that the additional improvement in fit in our preferred parameterization depicted by the red dotted line results from the combination of a high Kimball curvature and a lower Phillips curve slope, which when combined resolve the missing deflation puzzle. Finally, note that both high curvature parameterizations track inflation closely without showing any deterioration in GDP fit, while both low curvature parameterizations trigger large inflation declines during the recession and struggle to match actual GDP growth.

5.3. Implications for the log-likelihood

Having shown that a nonlinear formulation of the SW model can successfully resolve the missing deflation puzzle, we now explore the implications for the likelihood function. Before turning to the results, it is important to notice that price and wage markup shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ are introduced in the nonlinear model so that after log-linearization the model dynamics in response to these shocks are identical to those in the estimated linearized SW model. Equations (B.26) and (B.22) in the Technical Appendix show that $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ shift marginal costs and the marginal rate of substitution, respectively. Inflation dynamics in the nonlinear model close to the steady state evolve according to:

$$\hat{\pi}_t - \iota_p \hat{\pi}_{t-1} = \beta (E_t \hat{\pi}_{t+1} - \iota_p \hat{\pi}_t) + \kappa (\hat{m}\bar{c}_t + \hat{\varepsilon}_t^p), \quad (17)$$

By comparing Eq. (17) with the estimated linearized SW model (Eq. (16)), we see that the shock $\hat{\varepsilon}_t^p$ affects inflation isomorphically to $\hat{\varepsilon}_t^p$ up to a scaling factor κ . However, an important insight from the nonlinear model is that the kinked demand schedule effectively implies that κ is time-varying: κ decreases (increases) in recessions (booms). Accordingly, a given sized innovation to $\hat{\varepsilon}_t^p$ affects inflation less (more) in a recession (boom) relative to the linearized SW model in which the effects of an innovation to $\hat{\varepsilon}_t^p$ is time-invariant.

This feature implies that the nonlinear model, despite its ability to closely track inflation dynamics during the Great Recession and its aftermath, still relies on sizeable markup shocks to account for all (high-frequency) movements in price (and wage) inflation. Price markup shocks, for example, have to work their way through the marginal cost channel which is muted in deep recessions due to the nonlinear price Phillips curve. Accordingly, it is not evident that the nonlinear model will be associated with an improvement in the log-likelihood relative to the baseline SW model during the crisis.

We therefore also examine an additional specification of the nonlinear model in which the markup shocks are introduced in the nonlinear price and wage setting equations in a way that the transmission of these shocks to price and wage inflation is invariant to the state of the economy. Specifically, price and wage shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ are introduced in the nonlinear price and wage setting equations (see equations (B.25)–(B.27) and equations (B.21)–(B.23) in the Technical Appendix) in such a way that $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ are up to a first-order approximation and scaling observationally equivalent to the price and wage markup shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ without the need to work their way through the marginal cost channels anymore like $\tilde{\varepsilon}_t^p$ and

Table 2
Log-likelihood evaluation results.

	Markup shocks		Inflation shocks	
	SW	High curvature	SW	High curvature
Nonlinear model				
1965Q1–2007Q4	6400.98	6551.72	6399.19	6502.58
2008Q1–2014Q2	926.57	944.86	938.59	972.33
Linearized model	SW	High Curvature	SW	High Curvature
1965Q1–2007Q4	6403.42	6557.27	6403.42	6557.27
2008Q1–2014Q2	938.98	969.14	938.98	969.14

Notes: Columns labeled SW refer to the Smets and Wouters model calibrated at the mode of their original estimates, while columns labeled High Curvature refer to the Smets and Wouters model calibrated at the mode of our preferred re-estimated specification over the period 1965Q1–2007Q4. The log-likelihood for the linearized and nonlinear models are calculated using Eq. (18) conditional on these parameterizations and the structural shocks obtained using the filtering approach described in Section 5.1. Value in bold shows highest log-likelihood in the Great Recession sample period.

$\tilde{\varepsilon}_t^w$ have to do. Importantly, when linearizing the nonlinear model with these shocks, we obtain the same linearized price Phillips curve as in SW (see Eq. (16)). A drawback with this alternative specification of the nonlinear model is that the structural interpretation of the shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ is perhaps somewhat less clear (beyond that they imply deviations from profit maximizing prices and wages, hence a deviation from optimal markups). In Table 2, we will refer to the results based on $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ shocks as “Inflation shocks” compared to “Markup shocks” when $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ are used. Reporting log-likelihood results for the nonlinear model for both the markup shocks and inflation shocks cases allow us to parse out the impact on the likelihood of this model feature. Moreover, comparing the log-likelihood for the nonlinear specification with the inflation shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ with the linearized model allows us parse out the likelihood implications of intrinsic nonlinearities as the markup shocks $\tilde{\varepsilon}_t^p$ and $\tilde{\varepsilon}_t^w$ enter observationally equivalent in both models.

With this in mind, Table 2 compares the likelihood implications between the linearized and nonlinear solutions for the SW parametrization with our preferred re-estimated parameterization with a lower gross markup and higher Kimball parameter. The log-likelihood is computed following Fair and Taylor (1983),

$$L = \sum_{t=1}^T \log |J_t| - \frac{T}{2} \log |\Sigma| - \frac{1}{2} \sum_{t=1}^T \varepsilon_t' \Sigma^{-1} \varepsilon_t, \quad (18)$$

where J_t is the Jacobian of the model, Σ is the variance-covariance matrix and ε_t is the vector of filtered structural shocks.²¹ Hence, given a parametrization of the model, the solution and filtering procedure described above provides us with all the necessary elements in Eq. (18) to compute the likelihood.

In Table 2 we see that the log-likelihood is higher in the nonlinear model for our alternative parameterization than the SW parameterization under the markup shock interpretation. As expected, the likelihood for the re-estimated linearized specification is higher than the nonlinear model as the linearized model requires less volatile markup shocks than the nonlinear model. When these shocks are introduced so they remain additive in the Phillips curves in all states (i.e. using ‘Price and wage inflation shocks’), the likelihood of the nonlinear model increases significantly. Importantly, the nonlinear model now requires notably less volatile exogenous impulses to inflation dynamics, and in this case the log-likelihood in the nonlinear model with a high Kimball parameter is higher than in the linearized SW model during the crisis. Since the parameters in the nonlinear formulation have not been estimated to enhance its fit to the data, we take this result as suggestive of strong likelihood support for the nonlinear framework.

5.4. Accounting for skewness in the data

We have documented that the nonlinear model can account for the dynamics of inflation during the Great Recession and its aftermath, including the flatter slope of the Phillips curve. We now turn to stochastic simulations to examine whether the nonlinear model may also help to explain unconditional moments of price and wage inflation. Inflation data has a positive skew during the postwar period, i.e. there are episodes with inflation bursts and then there are episodes with very low and moderate rates of inflation but no long-lived deflationary episodes. This positive skew is a robust finding for different measures of inflation. The gray areas in Fig. 5 show the kernel smoothed distribution for price and wage inflation measured by the GDP deflator, hourly compensation and hourly earnings for the sample period 1965Q1–2014Q2.

The blue and red lines show the corresponding kernel smoothed distributions based on the stochastic simulations of the linearized and nonlinear model with our alternative parametrization of the model. They are based on a simulated sample of 10,000 observations, with the variance of the shocks set equal to the estimated variance from the linearized model in

²¹ This expression gives the “unconcentrated” log-likelihood up to a constant (see equation (20) in Fair and Taylor (1983)).

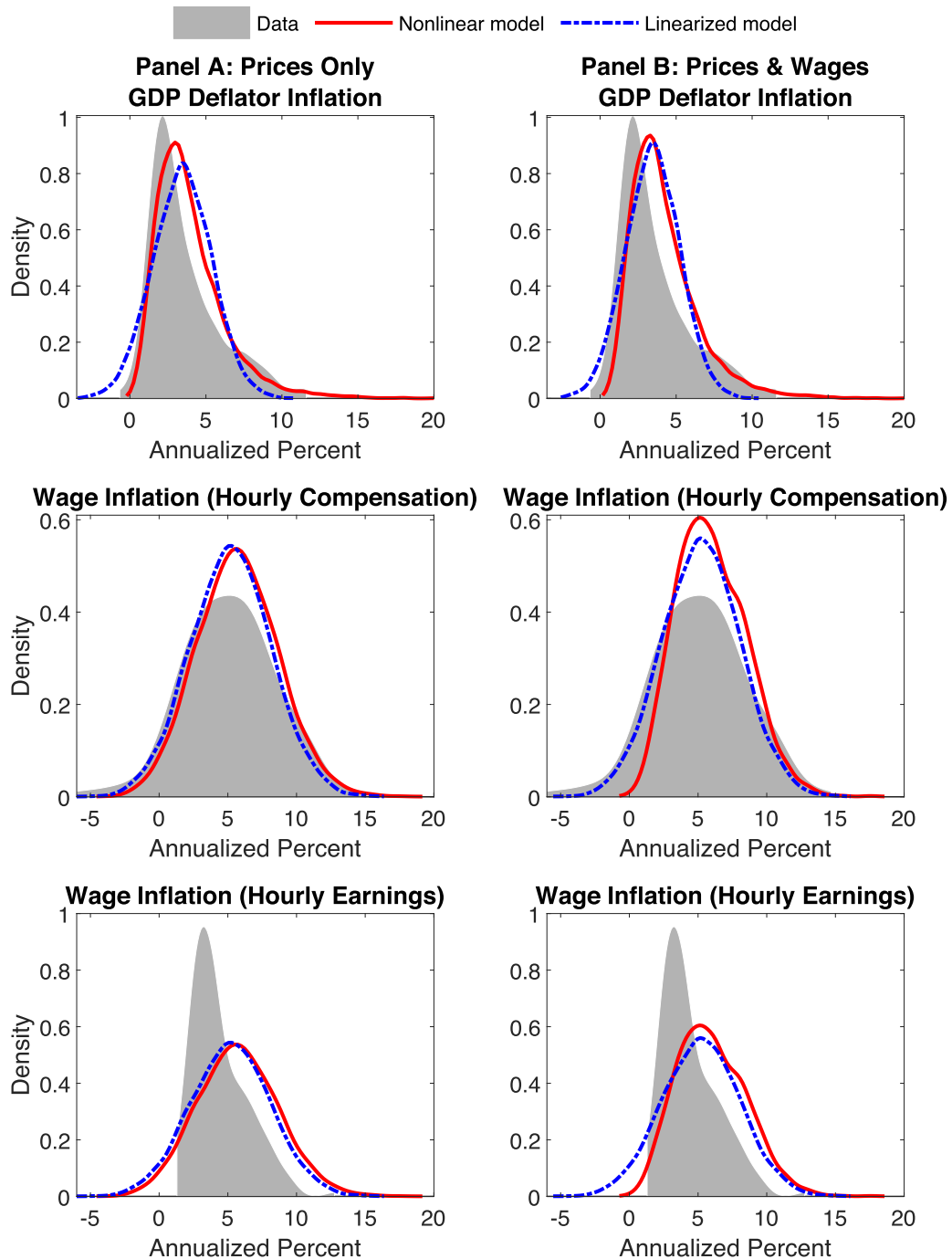


Fig. 5. Densities of data vs. stochastic model simulations: high demand curvature in prices only vs. prices & wages.

Section 5.2. As can be seen from the figure, the nonlinear model fits the unconditional statistical properties of inflation well – better than the linearized model. The linearized model features a symmetric normal distribution for inflation, with significant mass in deflationary territory. By contrast, the nonlinear solution features nearly zero density in deflationary territory for inflation, in line with the data. It is striking how much better the nonlinear model captures unconditional U.S. inflation dynamics during the post-war period compared to the linearized model.²²

²² Keep in mind that the model estimation has not used information about the skewness of inflation. In this sense, the results in Fig. 5 can be interpreted as a posterior predictive check.

Fig. 5 also depicts density plots for two alternative measures for nominal wage inflation: the middle row shows hourly compensation (the data used by SW when estimating the model) and the bottom row shows hourly earnings. An important difference between the two time series is that the hourly compensation series has considerable mass below nil and so the linearized and nonlinear SW models fits the empirical distribution about equally well. However, the wage series measured by hourly earnings in the bottom row displays a noticeable positive skewness consistent with downward nominal wage rigidity (DNWR), which neither the linearized nor the nonlinear model in Panel A can account for. However, if we re-calibrate the model to feature a higher Kimball parameter in wage setting and a lower gross steady state wage markup, the nonlinear solution captures the skewness of nominal wage inflation as measured by hourly earnings much better than the linearized model.²³ Effectively, this parameterization represents a micro-founded model of DNWR.

6. Conclusions

Formulating a nonlinear macroeconomic model goes a long way towards accounting for the missing deflation puzzle, i.e. the empirical fact that inflation fell little against the backdrop of a large and persistent fall in output during the Great Recession. Our resolution of the puzzle stresses the nonlinear influence of strategic complementarities and real rigidities in price- and wage-setting. Our proposed nonlinear framework is also attractive along the following additional dimensions: (i) it mitigates the tension between the macroeconomic evidence of a low Phillips curve slope and the microeconomic evidence of frequent price changes; (ii) it explains inflation dynamics prior to the crisis and the empirical positive skew in inflation without relying on a similarly positive skew in output (which would be counterfactual); and (iii) it sheds light on the missing inflation puzzle, i.e. why price and wage inflation remained persistently low in the aftermath of the Great Recession.

While our nonlinear model allows us to simultaneously explain the quantitative inflation performance prior to the Great Recession, the missing deflation during the Great Recession, as well as the missing inflation in recent years, a linearized version of the underlying nonlinear model fails to do so. All told, our results caution against the common practice of using linearized models to study inflation and output dynamics when the economy is exposed to large shocks.

Recent work by Aruoba et al. (2018); Gust et al. (2017) and Aruoba et al. (2020) have made tremendous progress in developing methods and algorithms to filter and estimate nonlinear DSGE models. Despite this progress, fully fledged global solution and nonlinear estimation of the SW model – even absent Kimball kinked demand curves in price and wage setting – is hardly feasible as it contains a large number of shocks and state variables. Adding Kimball (1995) aggregation complicates matters even more as it increases the number of endogenous state variables due to various price and wage dispersion terms. Thus, a fully fledged global solution and estimation exercise of the nonlinear SW model (including Kimball aggregation) is infeasible at this point in time and we leave progress along this dimension to future work.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jmoneco.2021.09.003](https://doi.org/10.1016/j.jmoneco.2021.09.003)

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²³ Specifically, the model in Panel B sets $\epsilon_w = 40$ instead of $\epsilon_w = 10$ and $\phi_w = 1.1$ instead of $\phi_w = 1.5$ (as in the baseline SW model). As with prices, this alternative parameterization has a somewhat smaller sensitivity of wage inflation to the wage markup wedge in the linearized wage Phillips curve.

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